

Midterm Activity #3 - Basic ATM System

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Course, Year & Section: BSIT 2-3

Course Code & Description: COMP 009 Object Oriented Programming

Submitted to: Sir SEVERINO M. BEDIS JR.

General Instructions

Create a Java program that accepts student information and evaluates the student's grade

Using decision structures.

Program Requirements

1. Create a Java program named BasicATMSystemLastName.
2. Use the Scanner class to accept user input.
3. Assume the initial balance is 5,000 pesos.

Display the Menu

===== ATM MENU =====

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Program Rules

- Use a switch statement to process the selected option.
- Use if-else statements to validate withdrawal.
- Maximum withdrawal per transaction: 2,000 pesos.
- The program must repeat using a do-while loop until the user selects Exit.

Submission Format

- Submit the Java source code file (.java).
- The filename must follow this format:
Section_LastName_FirstName_ActivityName.java
- Include screenshots of the program output showing the following:
 - User input
 - Program processing
 - Final output/result
- The screenshot must clearly display the entire console output.

PART 1 of the screenshot User Input

The screenshot displays the Programiz Online Java Compiler interface. The editor shows the source code for `BasicATMSysAvalo.java`, which includes a menu-driven ATM system. The output window shows the program's execution, displaying the menu and the first choice entered by the user.

```

1 import java.util.Scanner;
2
3 public class BasicATMSysAvalo {
4
5     public static void displayprogram() {
6         System.out.println("!!-----BASIC ATM SYSTEM ni JAY-----!!");
7     }
8
9     public static void displayFinancialSummary(int transactionAmount, int
    remainblnc, String statesg) {
10        System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!");
11        System.out.printf(" Transaction Amount: %d pesos\n", transactionAmount
    );
12        System.out.printf(" Current Balance: %d pesos\n", remainblnc);
13        System.out.println("-----");
14        System.out.println(" Status: " + statesg);
15        System.out.println("-----");
16    }
17
18    public static void main(String[] args) {
19        Scanner scanner = new Scanner(System.in);
20        int choice = 0;
21        int remainblnc = 5000;
22
23        displayprogram();
24
25        do {
26            System.out.println("\n==== ATM MENU =====");
27            System.out.println("1. Check Balance");
28            System.out.println("2. Deposit");
29            System.out.println("3. Withdraw");
30            System.out.println("4. Exit");
31            System.out.print("Please enter your choice: ");
32
33            if (scanner.hasNextInt()) {

```

```

!!-----BASIC ATM SYSTEM ni JAY-----!!

==== ATM MENU ====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 1

!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 0 pesos
Current Balance: 5000 pesos
-----
Status: Great job u Da Man! Balance checked successfully.
=====

==== ATM MENU ====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice:

```

PART 2 of the screenshot User Input/progress

```

1 import java.util.Scanner;
2
3 public class BasicATMSysArevalo {
4
5     public static void displayprogram() {
6         System.out.println("!!-----BASIC ATM SYSTEM ni JAY-----!!");
7     }
8
9     public static void displayFinancialSummary(int transactionAmount, int
    remainblnc, String statmsg) {
10        System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
11        System.out.printf(" Transaction Amount: %d pesos\n", transactionAmount
        );
12        System.out.printf(" Current Balance: %d pesos\n", remainblnc);
13        System.out.println("-----");
14        System.out.println(" Status: " + statmsg);
15        System.out.println("=====");
16    }
17
18    public static void main(String[] args) {
19        Scanner scanner = new Scanner(System.in);
20        int choice = 0;
21        int remainblnc = 5000;
22
23        displayprogram();
24
25        do {
26            System.out.println("\n===== ATM MENU =====");
27            System.out.println("1. Check Balance");
28            System.out.println("2. Deposit");
29            System.out.println("3. Withdraw");
30            System.out.println("4. Exit");
31            System.out.print("Please enter your choice: ");
32
33            if (scanner.hasNextInt()) {

```

```

!!-----BASIC ATM SYSTEM ni JAY-----!!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 2
Enter deposit amount (whole numbers only BAWAL MAY CENTAVOS): 500.50
--> Invalid: Please enter an actual numeric value (whole numbers only hindi ung butas).
Enter deposit amount (whole numbers only BAWAL MAY CENTAVOS): 500

!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 500 pesos
Current Balance: 5500 pesos

-----
Status: Deposit successful! You managed your finances well.
=====

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: |

```

PART 3 of the screenshot User Input/progress

```

1 import java.util.Scanner;
2
3 public class BasicATMSysArevalo {
4
5     public static void displayprogram() {
6         System.out.println("!!-----BASIC ATM SYSTEM ni JAY-----!!");
7     }
8
9     public static void displayFinancialSummary(int transactionAmount, int
    remainblnc, String statmsg) {
10        System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
11        System.out.printf(" Transaction Amount: %d pesos\n", transactionAmount
        );
12        System.out.printf(" Current Balance: %d pesos\n", remainblnc);
13        System.out.println("-----");
14        System.out.println(" Status: " + statmsg);
15        System.out.println("=====");
16    }
17
18    public static void main(String[] args) {
19        Scanner scanner = new Scanner(System.in);
20        int choice = 0;
21        int remainblnc = 5000;
22
23        displayprogram();
24
25        do {
26            System.out.println("\n===== ATM MENU =====");
27            System.out.println("1. Check Balance");
28            System.out.println("2. Deposit");
29            System.out.println("3. Withdraw");
30            System.out.println("4. Exit");
31            System.out.print("Please enter your choice: ");
32
33            if (scanner.hasNextInt()) {

```

```

!!-----BASIC ATM SYSTEM ni JAY-----!!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 3
Enter withdrawal amount (whole numbers only): 500.50
--> Invalid: Please enter an actual numeric value (whole numbers only hindi ung butas).
Enter withdrawal amount (whole numbers only): 500

!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 500 pesos
Current Balance: 4500 pesos

-----
Status: Withdrawal successful! Enjoy your scatter cash GOAT.
=====

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice:

```

PART 3 of the screenshot User Input/progress

```

11
12 public static String checkBdgtStatus(String name, double totalEXPN, double
    bdgt) {
13     if (totalEXPN > bdgt) {
14         return "WARNING!, " + name + "! HEY! Yes you! You have exceeded your
            budget. Time to cut back!";
15     } else if (totalEXPN == bdgt) {
16         return "OOPS! Careful Goat, " + name + ". You reached your budget
            limit. Utang next time!";
17     } else {
18         return "Great job u Da Man!, " + name + "! You managed your finances
            well and stayed under budget.";
19     }
20 }
21
22 public static void displayFinancialSummary(double totalEXPN, double bdgt,
    double remainingBalance, String statmesg) {
23     System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
24     System.out.printf(" Total bdgt: $%.2f\n", bdgt);
25     System.out.printf(" Total Expenses: $%.2f\n", totalEXPN);
26     System.out.printf(" Remaining Balance: $%.2f\n", remainingBalance);
27     System.out.println("-----");
28     System.out.println(" Status: " + statmesg);
29     System.out.println("=====");
30 }
31
32 public static void main(String[] args) {
33     Scanner scanner = new Scanner(System.in);
34     String choice;
35
36     do {
37         displayProgramTitle();
38         System.out.print("Please enter your name: ");
39         String name = scanner.nextLine();
40         double bdgt = getValidMonetaryInput(scanner, "Oke! Enter your total
            budget: $");
41
42         // ... (rest of the code)
43     } while (choice != "exit");
44 }

```

Output:

```

!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your names: Jay
Oke! Enter your total budget: $100
Enter your food expenses: $10
Enter transportation expenses: $10
Enter other expenses : $10
Enter your addition expenses ehem scatter: $1000

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $100.00
Total Expenses: $1030.00
Remaining Balance: $-930.00
-----
Status: WARNING!, Jay! HEY! Yes you! You have exceeded your budget. Time to cut back!
=====
Would you like to try again and track another person's expenses? (yes/no/n/exit): yes
!!-----PERSONAL EXPENSE TRACKER ni JAY-----!!
Please enter your name: Jay 2
Oke! Enter your total budget: $1000
Enter your food expenses: $10
Enter transportation expenses: $10
Enter other expenses : $10
Enter your addition expenses ehem scatter: $100

!!-----FINANCIAL SUMMARY MO PO!-----!!

Total bdgt: $1000.00
Total Expenses: $130.00
Remaining Balance: $870.00
-----
Status: Great job u Da Man!, Jay 2! You managed your finances well and stayed under
budget.
=====

```

PART 4 of the screenshot Results

```

1 import java.util.Scanner;
2
3 public class BasicATMSysTemArevalo {
4
5     public static void displayprogram() {
6         System.out.println("!!-----BASIC ATM SYSTEM ni JAY-----!!");
7     }
8
9     public static void displayFinancialSummary(int transactionAmount, int
    remainblnc, String statmesg) {
10        System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
11        System.out.printf(" Transaction Amount: %d pesos\n", transactionAmount
        );
12        System.out.printf(" Current Balance: %d pesos\n", remainblnc);
13        System.out.println("-----");
14        System.out.println(" Status: " + statmesg);
15        System.out.println("=====");
16    }
17
18    public static void main(String[] args) {
19        Scanner scanner = new Scanner(System.in);
20        int choice = 0;
21        int remainblnc = 5000;
22
23        displayprogram();
24
25        do {
26            System.out.println("\n==== ATM MENU =====");
27            System.out.println("1. Check Balance");
28            System.out.println("2. Deposit");
29            System.out.println("3. Withdraw");
30            System.out.println("4. Exit");
31            System.out.print("Please enter your choice: ");
32
33            if (scanner.hasNextInt()) {
34                // ... (rest of the code)
35            }
36        } while (choice != 4);
37    }
38 }

```

Output:

```

!!-----BASIC ATM SYSTEM ni JAY-----!!

==== ATM MENU ====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 4
Thank you for using my very own Basic ATM System. Goodbye Alien X!!!

*** Code Execution Successful ***

```

PART 5 of the screenshot Results

Output

```
!!-----BASIC ATM SYSTEM ni JAY-----!!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 2
Enter deposit amount (whole numbers only BAWAL MAY CENTAVOS): 500

!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 500 pesos
Current Balance: 5500 pesos
-----
Status: Deposit successful! You managed your finances well.
=====

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 3
Enter withdrawal amount (whole numbers only): 500

!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 500 pesos
Current Balance: 5000 pesos
-----
Status: Withdrawal successful! Enjoy your scatter cash GOAT.
=====
```

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 1

!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 0 pesos
Current Balance: 5000 pesos
-----
Status: Great job u Da Man! Balance checked successfully.
=====

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice:
```

PART 6 of the screenshot Results of the max withdrawal and exit

Output

Clear

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 2
Enter deposit amount (whole numbers only BAWAL MAY CENTAVOS): 22000
```

```
!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 22000 pesos
Current Balance: 27000 pesos
-----
Status: Deposit successful! You managed your finances well.
=====
```

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 3
Enter withdrawal amount (whole numbers only): 2200
```

```
!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 2200 pesos
Current Balance: 27000 pesos
-----
Status: WARNING!, HEY! Yes you! Max withdrawal per transaction is 2,000 pesos. Time to cut back!
=====
```

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 3
Enter withdrawal amount (whole numbers only): 2200
```

```
!!-----FINANCIAL SUMMARY MO PO!-----!!

Transaction Amount: 2200 pesos
Current Balance: 27000 pesos
-----
Status: WARNING!, HEY! Yes you! Max withdrawal per transaction is 2,000 pesos. Time to cut back!
=====
```

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Please enter your choice: 4
Thank you for using my very own Basic ATM System. Goodbye Alien X!!!
```

```
=== Code Execution Successful ===
```

Code

```

import java.util.Scanner;

public class BasicATMSystemArevalo {

    public static void displayprogram() {
        System.out.println("!!-----BASIC ATM SYSTEM ni JAY-----!!");
    }

    public static void displayFinancialSummary(int transactionAmount, int remainblnc,
String statmesg) {
        System.out.println("\n!!-----FINANCIAL SUMMARY MO PO!-----!!\n");
        System.out.printf(" Transaction Amount: %d pesos\n", transactionAmount);
        System.out.printf(" Current Balance: %d pesos\n", remainblnc);
        System.out.println("-----");
        System.out.println(" Status: " + statmesg);
        System.out.println("=====\\n");
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int choice = 0;
        int remainblnc = 5000;

        displayprogram();

        do {
            System.out.println("\n===== ATM MENU =====");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit");
            System.out.println("3. Withdraw");
            System.out.println("4. Exit");
            System.out.print("Please enter your choice: ");

            if (scanner.hasNextInt()) {
                choice = scanner.nextInt();
                scanner.nextLine();

                switch (choice) {
                    case 1:
                        displayFinancialSummary(0, remainblnc, "Great job u Da Man!
Balance checked successfully.");
                        break;

                    case 2:
                        int depositAmount = getValidIntInput(scanner, "Enter deposit
amount (whole numbers only BAWAL MAY CENTAVOS): ");
                        if (depositAmount > 0) {
                            remainblnc += depositAmount;

```

```

        String statmesg = "Deposit successful! You managed your
finances well.";
        displayFinancialSummary(depositAmount, remainblnc,
statmesg);
    } else {
        System.out.println("--> Invalid man: Deposit amount must
be greater than 0.");
    }
    break;

    case 3:
        int withdrawamnt = getValidIntInput(scanner, "Enter withdrawal
amount (whole numbers only): ");
        String statmesg = "";

        if (withdrawamnt <= 0) {
            System.out.println("--> Invalid man: Withdrawal amount
must be greater than 0.");
        } else if (withdrawamnt > 2000) {
            statmesg = "WARNING!, HEY! Yes you! Max withdrawal per
transaction is 2,000 pesos. Time to cut back!";
            displayFinancialSummary(withdrawamnt, remainblnc,
statmesg);
        } else if (withdrawamnt > remainblnc) {
            statmesg = "OOPS! Careful Lebron. Insufficient funds.
Utang ka nalang next time!";
            displayFinancialSummary(withdrawamnt, remainblnc,
statmesg);
        } else {
            remainblnc -= withdrawamnt;
            statmesg = "Withdrawal successful! Enjoy your scatter cash
GOAT.";
            displayFinancialSummary(withdrawamnt, remainblnc,
statmesg);
        }
        break;

    case 4:
        System.out.println("Thank you for using my very own Basic ATM
System. Goodbye Alien X!!!");
        break;

    default:
        System.out.println("--> Invalid: man OR lady. But please
select a valid option from the menu.");
    }
    } else {
        System.out.println("--> Invalid! Ekis! Please enter an actual numeric
value from the menu.");
        scanner.next();
        scanner.nextLine();
    }
}

} while (choice != 4);

```

```
        scanner.close();
    }

    private static int getValidIntInput(Scanner scanner, String prompt) {
        int amount;
        while (true) {
            System.out.print(prompt);
            if (scanner.hasNextInt()) {
                amount = scanner.nextInt();
                scanner.nextLine();

                if (amount >= 0) {
                    break;
                } else {
                    System.out.println("--> Invalid man: Amount cannot be negative!
Try again.");
                }
            } else {
                System.out.println("--> Invalid: Please enter an actual numeric value
(whole numbers only hindi ung butas).");
                scanner.next();
                scanner.nextLine();
            }
        }
        return amount;
    }
}
```